

PRINCE MODI

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EDUCATION

University of California, San Diego

Sep 2025 – Jun 2027

MS Computer Science | GPA: 3.88/4.00

- Relevant Coursework: Distributed Systems, Deep Learning Systems, LLM System Optimization

Ganpat University, India

Aug 2018 – Jul 2022

B. Tech. Computer Engineering | GPA: 3.96/4.00

- Relevant Coursework: Operating Systems, Cloud Computing | **Academic Scholarship** | 2nd rank out of 60+ students

WORK EXPERIENCE

Picasso Lab, UC San Diego

Jul 2026 – Present

Graduate Student Researcher, LLM Inference Systems

- Profiling **attention-operator** compute and communication cost across prefill (varying prompt/KV-cache lengths) and decode (varying batch size/KV-cache lengths), fitting analytical regression models to measured latencies
- Building a **trace-driven simulator** that pairs the fitted cost models with multi-turn conversation traces to compare parallelism strategies (CP/TP/DP + EP) for a disaggregated prefill/decode deployment

Indian Institute of Science (IISc), Bangalore

Jan 2023–Apr 2024 (Full-time); Apr 2024–Nov 2024 (Voluntary)

Project Associate, Distributed Systems

Flotilla: Distributed Federated Learning Framework

- Architected a distributed training framework in Python/AsyncIO orchestrating **1,000+ worker nodes**, implementing both synchronous and asynchronous parameter aggregation strategies (FedAvg, FedAsync, tiered aggregation) via a pluggable event-driven state-machine design
- Designed a stateful parameter-server architecture with stateless workers, externalizing all orchestration state to **Redis** for real-time durability, cutting per-round overhead to just **1.7%** at 1,000+ nodes vs. **54.9%** for a leading open-source baseline
- Built **gRPC + MQTT** based worker discovery and heartbeat-driven liveness detection, and implemented server-side fault tolerance via incremental checkpointing and hot failover, recovering a live session on a standby server within **820ms** with rollback limited to a single round
- Validated the framework across heterogeneous accelerators (Nvidia Jetson GPUs, ARM edge devices) training CNN, LSTM, and transformer (ALBERT) architectures on PyTorch/HuggingFace, demonstrating tolerance to **~40% worker dropout** with negligible accuracy impact

Sterlite Technologies Limited, Ahmedabad

Jan 2022 – Aug 2022

Software Engineering Intern

- Refactored Docker images on Linux CI/CD pipelines, reducing image size by **35%** and accelerating build times by **50%**
- Integrated a Liquibase module to track MongoDB schema changes, improving version control reliability across environments

PUBLICATION

Roopkatha B., Prince Modi, et al. **Flotilla: A Scalable, Modular and Resilient Federated Learning Framework for Heterogeneous Resources**. *Journal of Parallel and Distributed Computing (JPDC)* (2025)

PROJECTS

Distributed Container Snapshotting | Podman, CRIU

June 2026

- Designed a distributed **snapshotting system** for stateful containerized applications, implementing the **Chandy-Lamport algorithm** to capture globally consistent cuts across nodes communicating over UDP
- Built a **reliable-UDP sidecar proxy** using sequence numbering, buffering, and acknowledgment retransmission to guarantee in-order, exactly-once delivery without modifying application code

FlashAttention Kernel | Triton, Python

Feb 2026

- Implementing **FlashAttention** and RMSNorm kernels in **Triton**, applying **tiled softmax** and **online normalization** to minimize HBM memory accesses and improve throughput for long-context LLM inference

TECHNICAL SKILLS

- Languages: Python (proficient), C, Java, Bash/Shell
- ML/LLM Frameworks: PyTorch, HuggingFace, SGLang, NCCL, Triton
- Distributed Systems: gRPC, MQTT, Redis, Fault-Tolerant Orchestration, Checkpoint/Restore
- Infrastructure: Linux, Docker/Podman, CRIU, PostgreSQL, Git